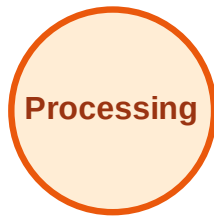


BFS Traversal

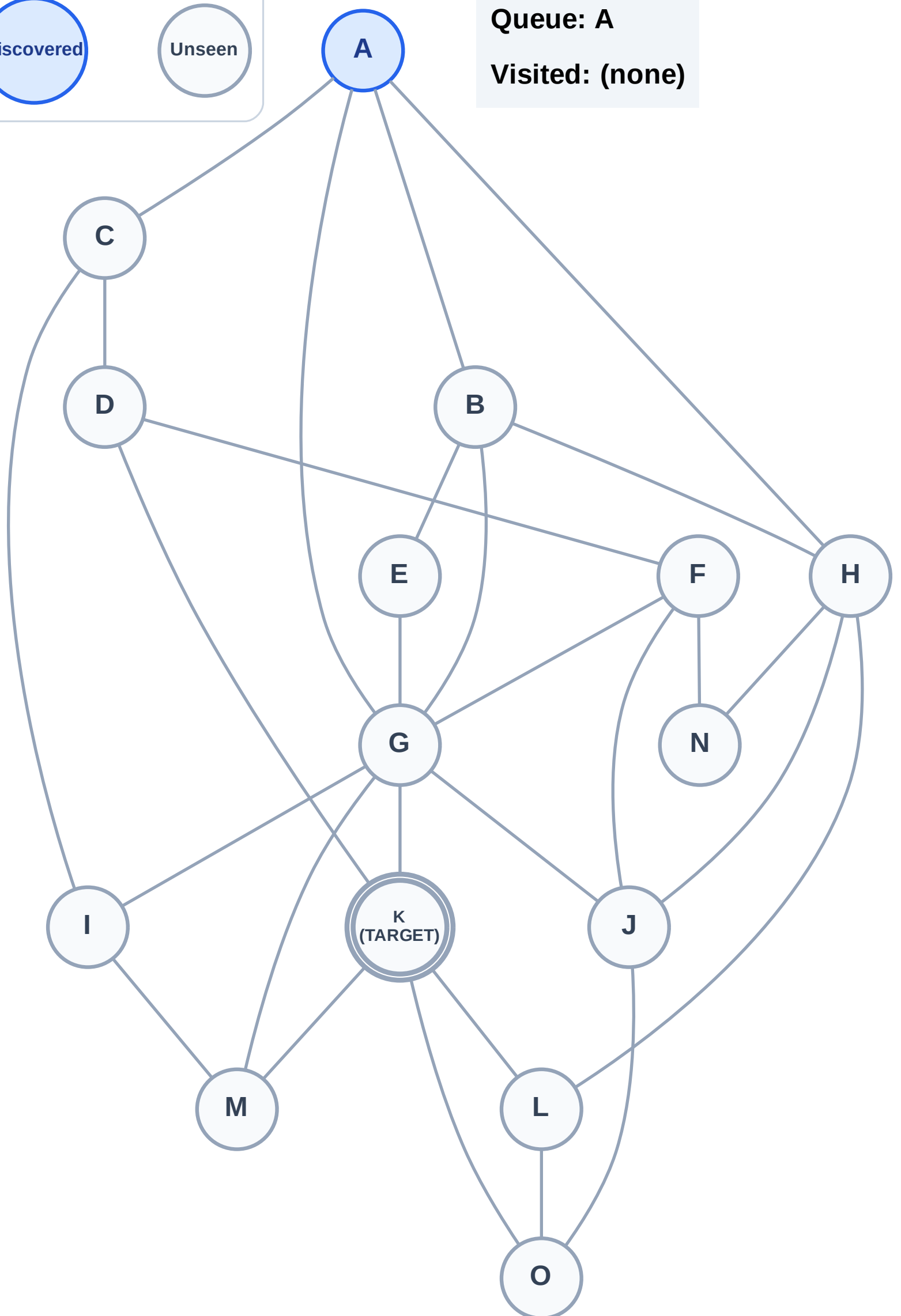
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

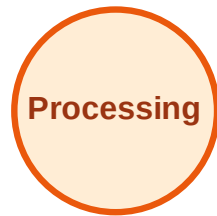
Visited: (none)



BFS Traversal

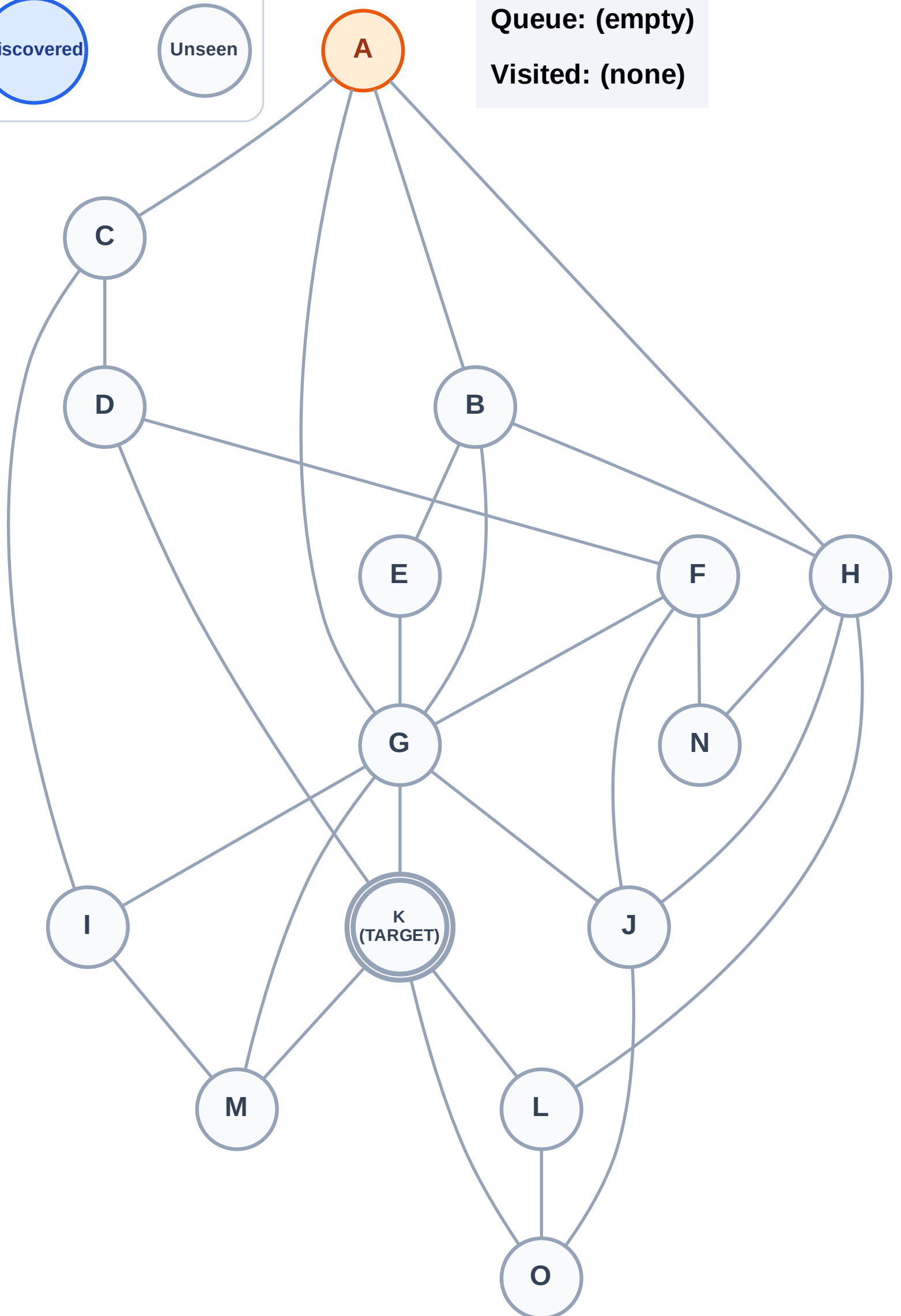
Step 2: Process node A

Legend



Queue: (empty)

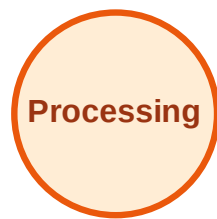
Visited: (none)



BFS Traversal

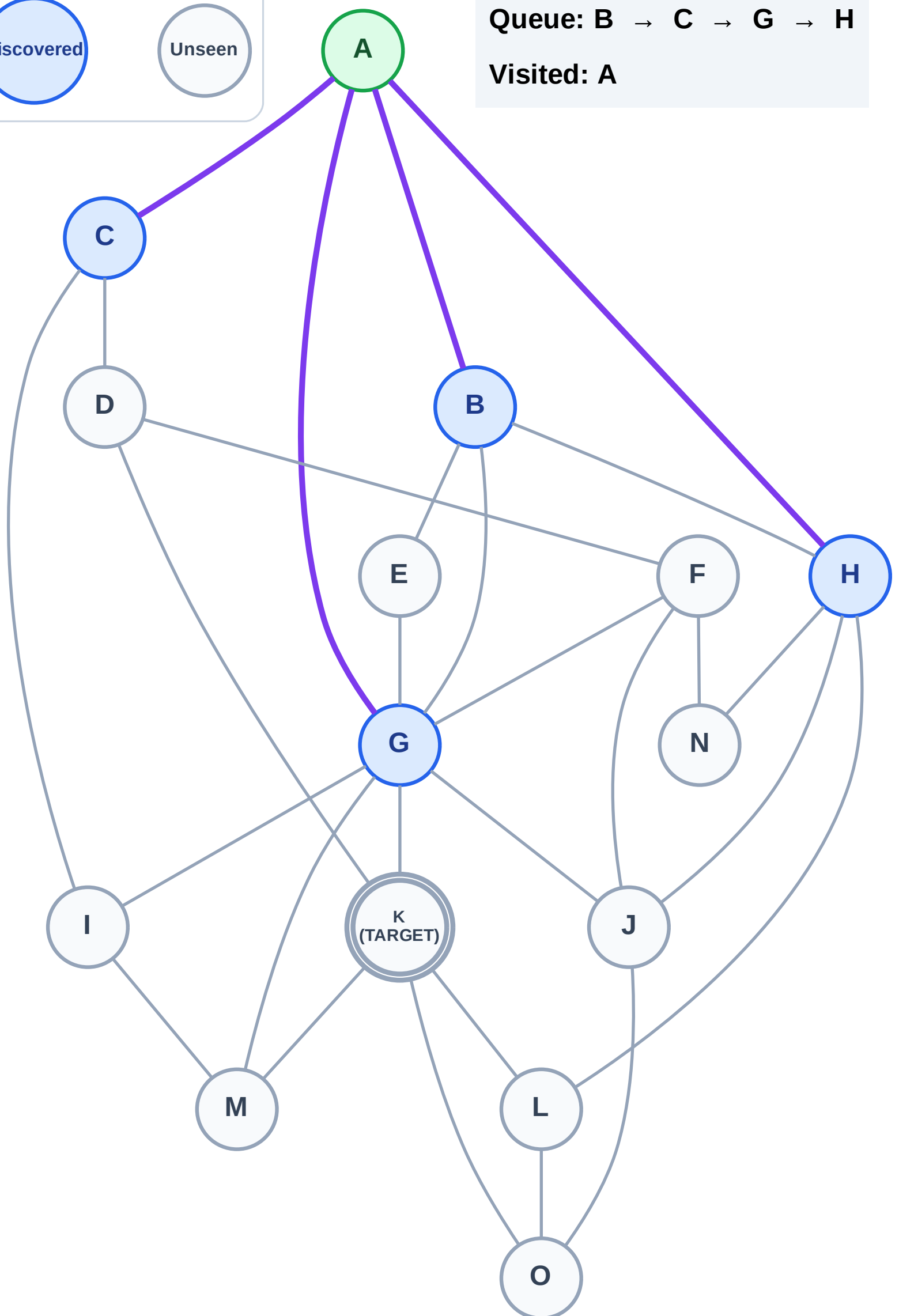
Step 3: Discover B, C, G, H from A

Legend



Queue: B → C → G → H

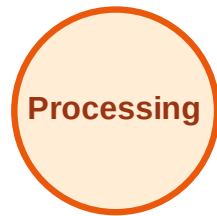
Visited: A



BFS Traversal

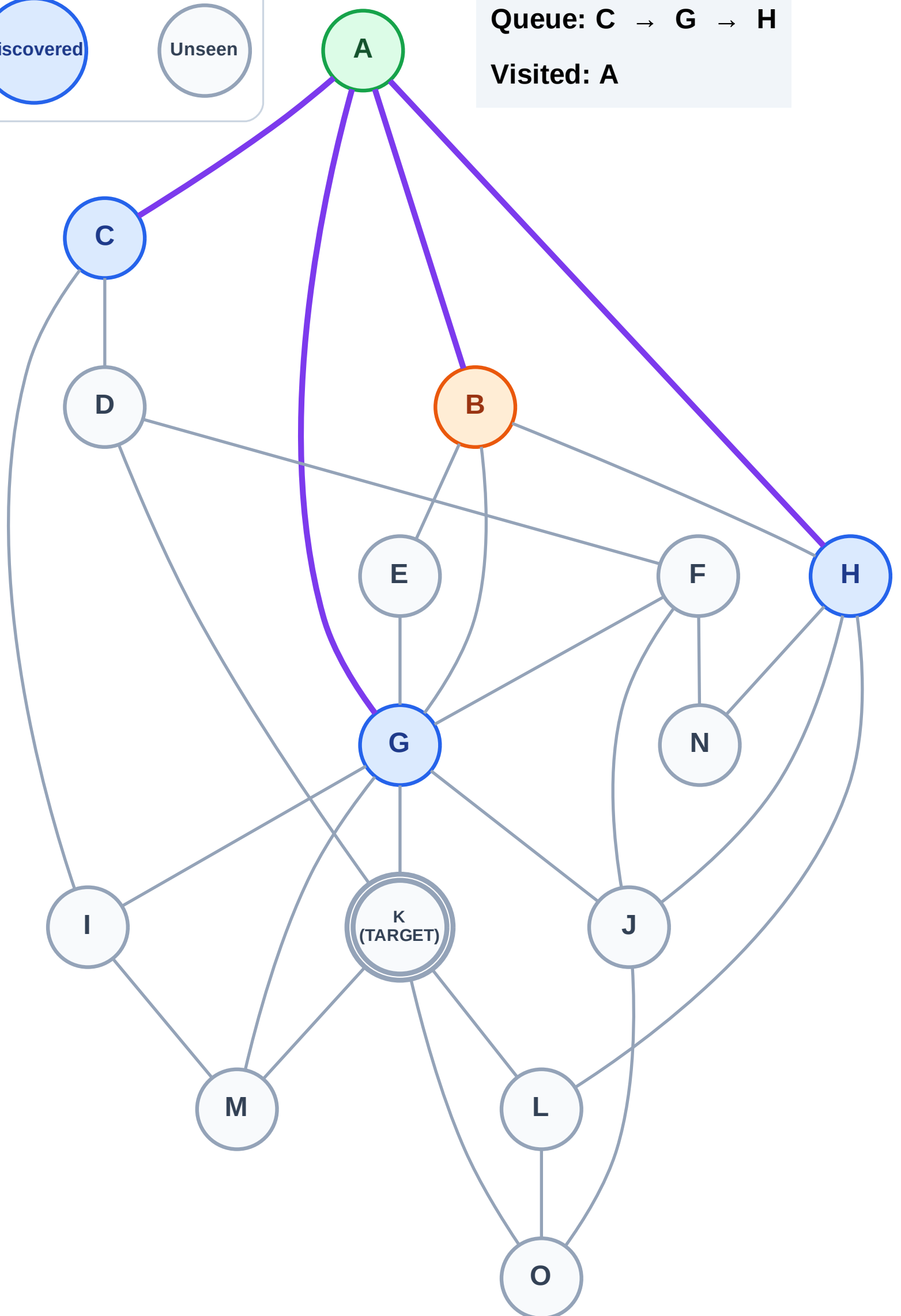
Step 4: Process node B

Legend



Queue: C → G → H

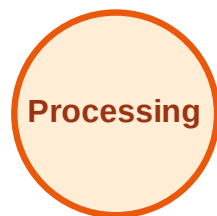
Visited: A



BFS Traversal

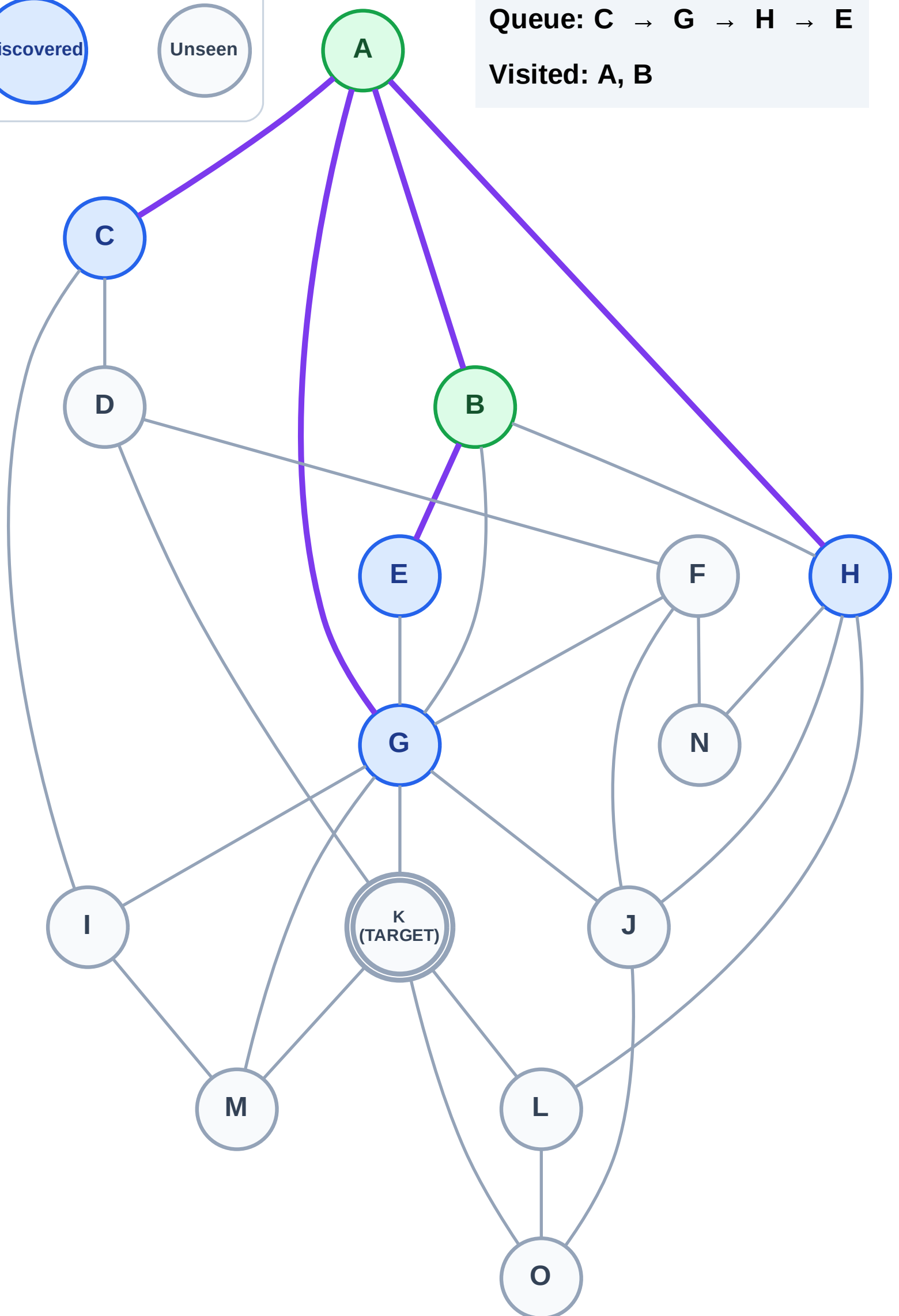
Step 5: Discover E from B

Legend



Queue: C → G → H → E

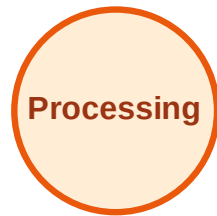
Visited: A, B



BFS Traversal

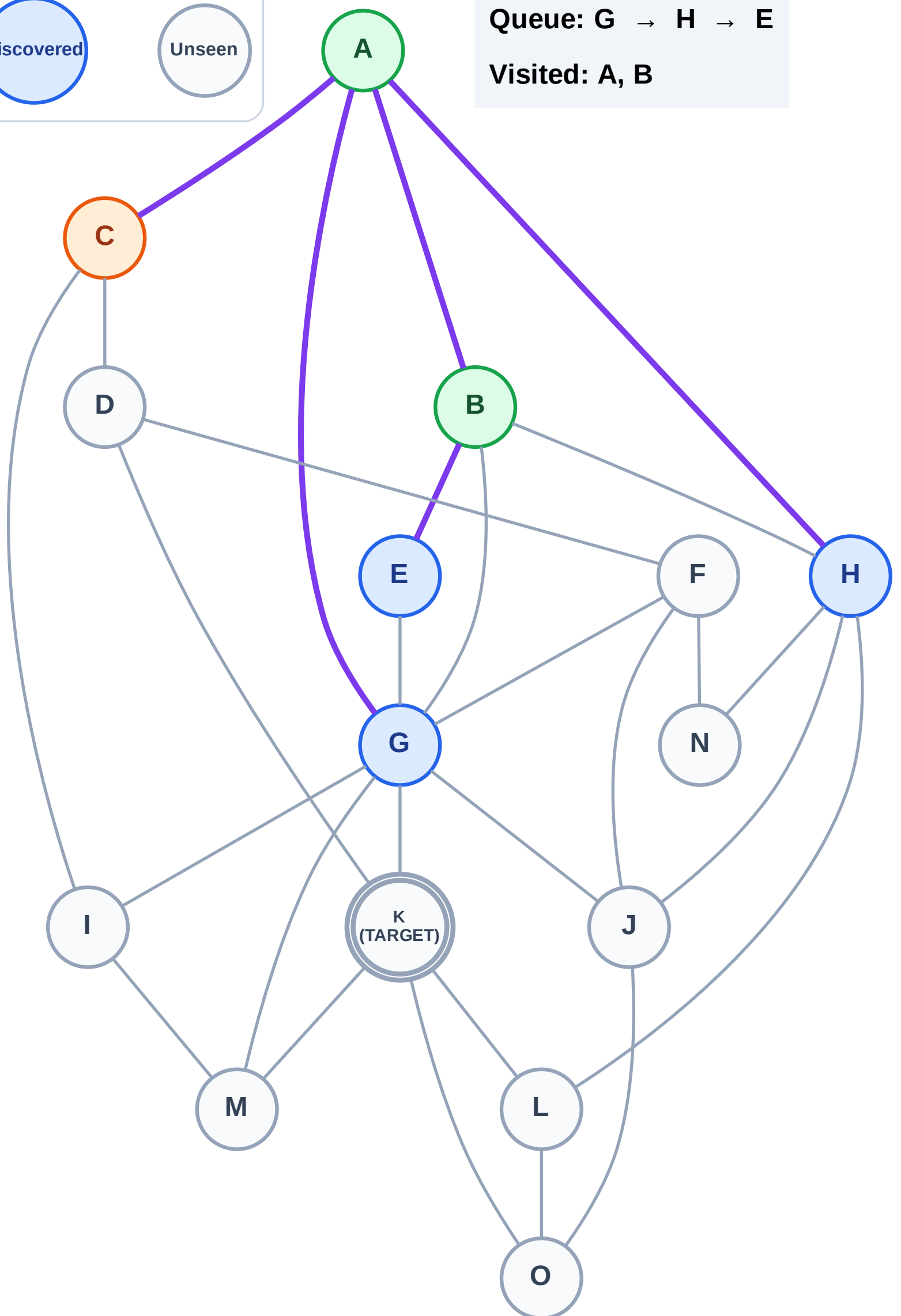
Step 6: Process node C

Legend



Queue: G → H → E

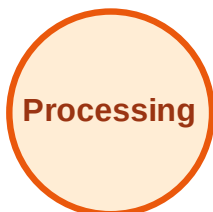
Visited: A, B



BFS Traversal

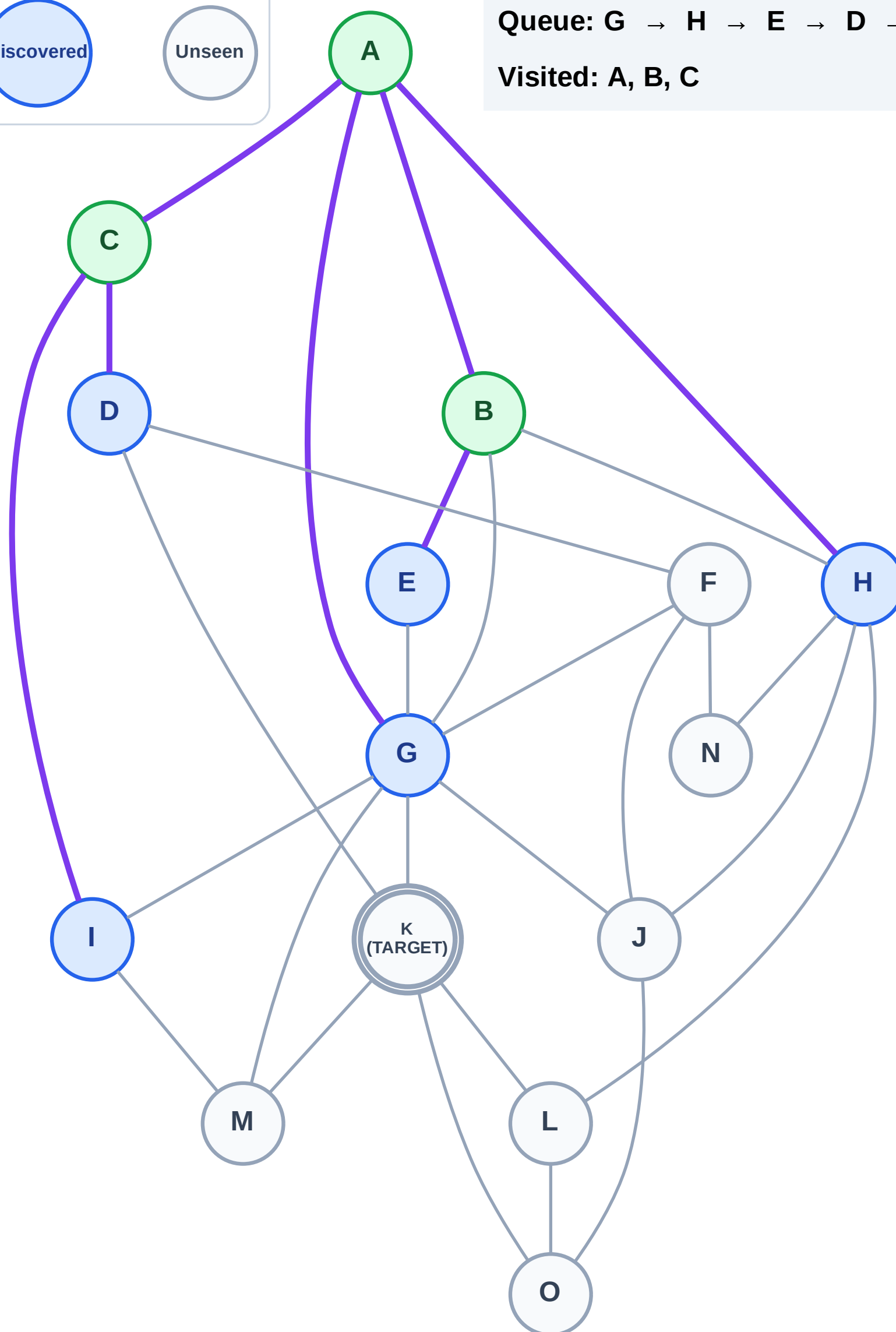
Step 7: Discover D, I from C

Legend



Queue: G → H → E → D → I

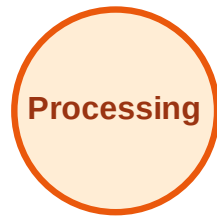
Visited: A, B, C



BFS Traversal

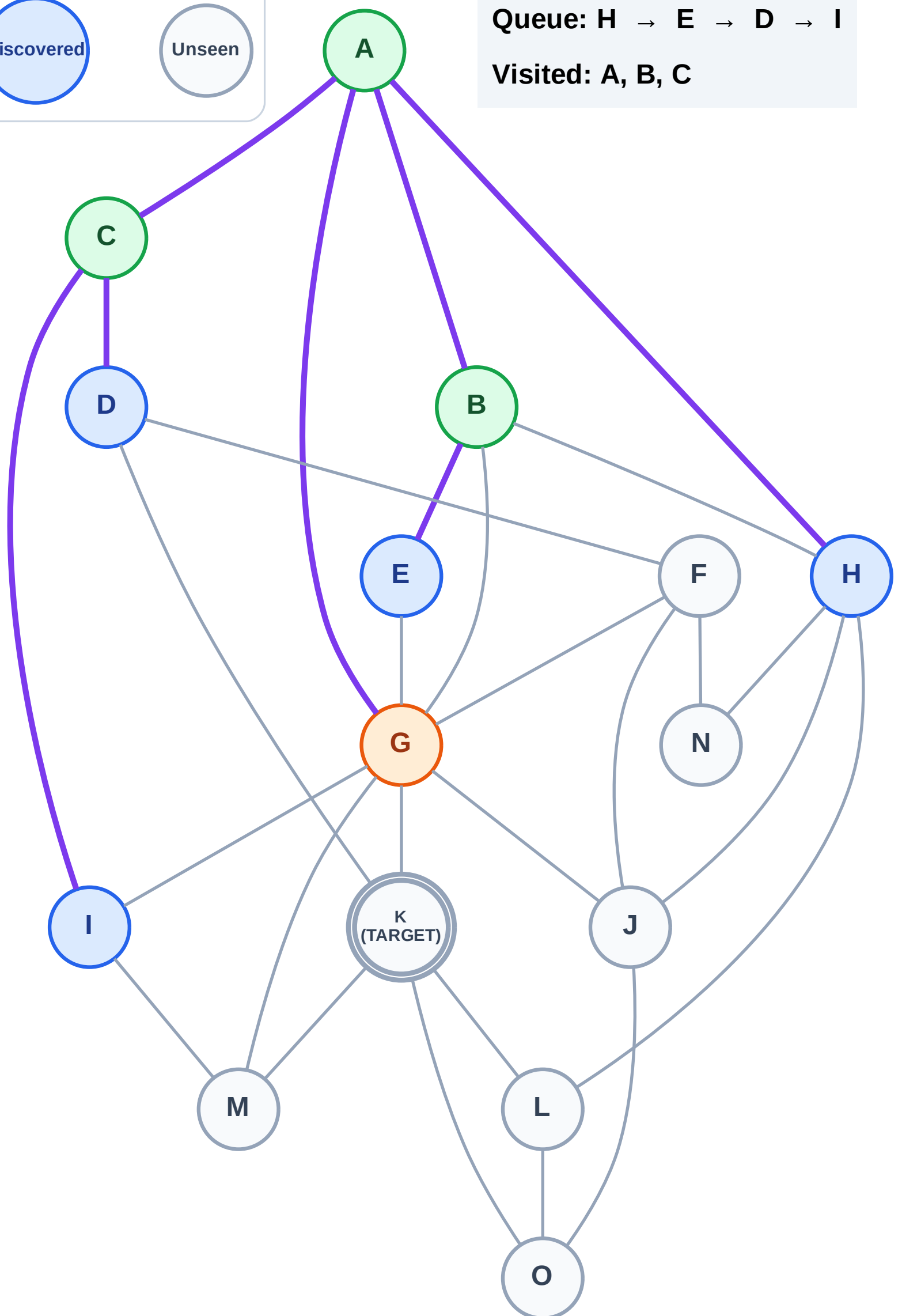
Step 8: Process node G

Legend



Queue: H → E → D → I

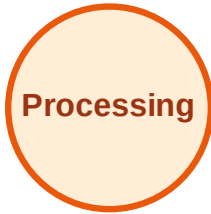
Visited: A, B, C



BFS Traversal

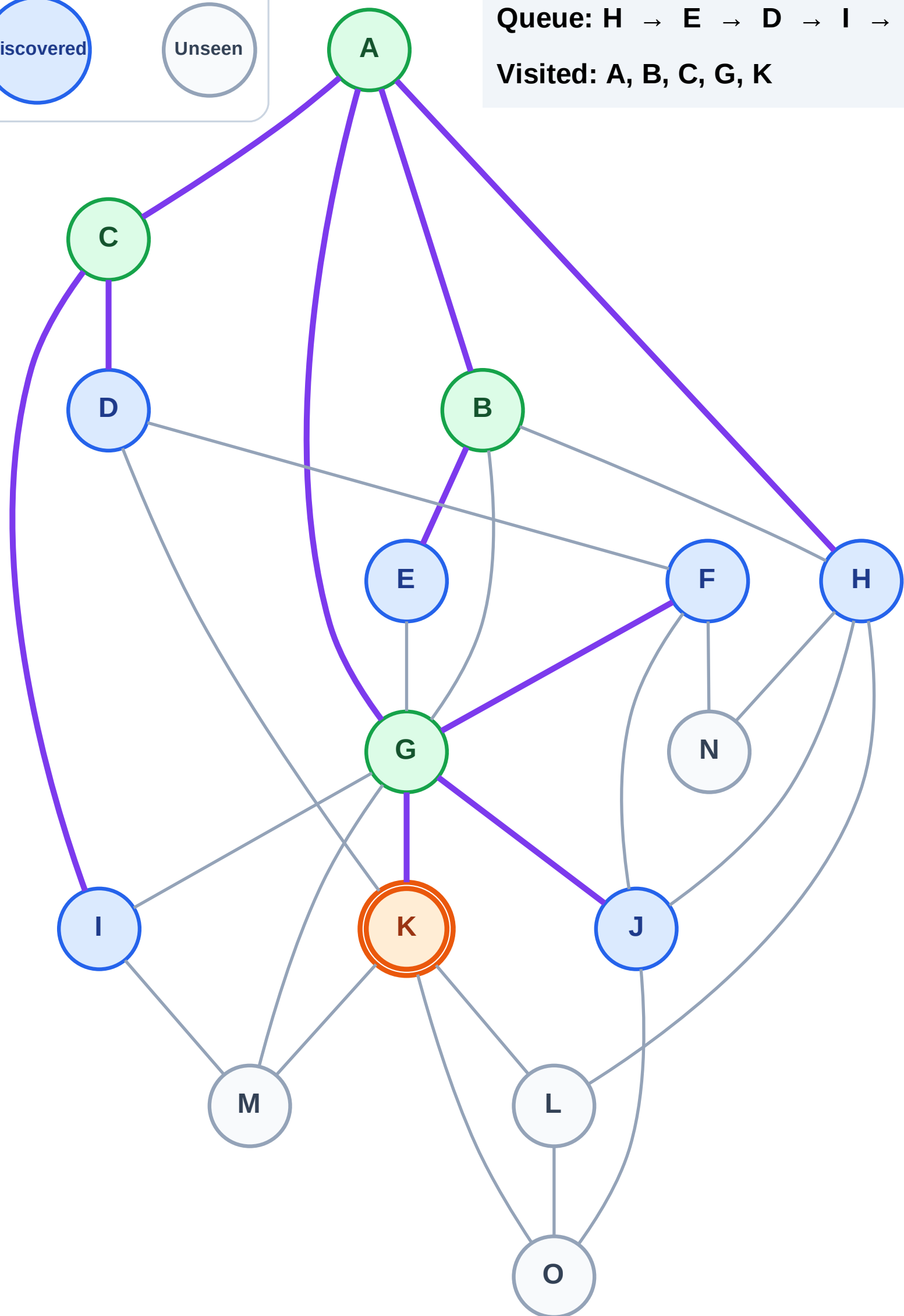
Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → G → K

Legend



Queue: H → E → D → I → F → J

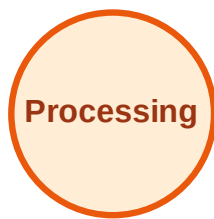
Visited: A, B, C, G, K



DFS Traversal

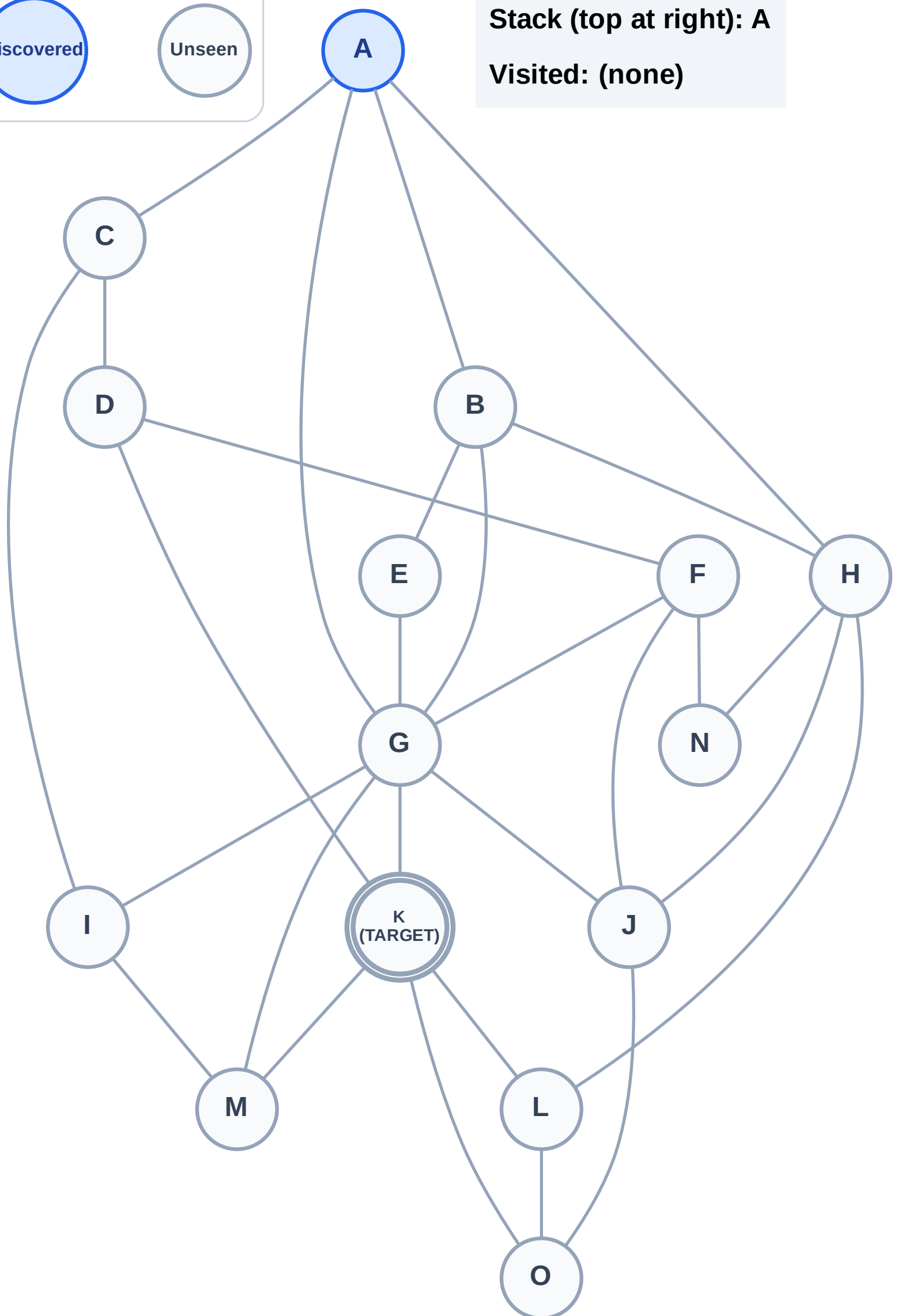
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

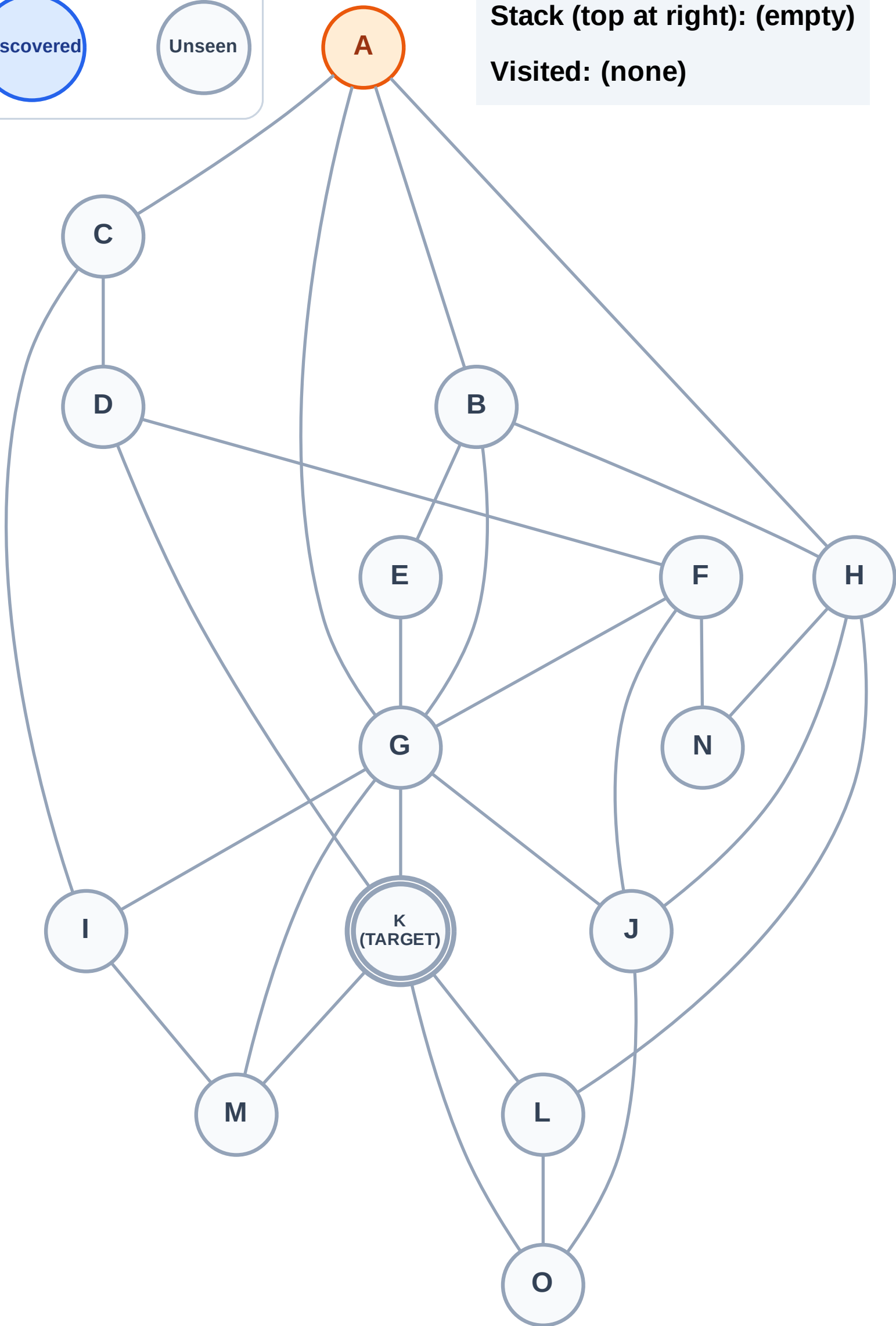
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

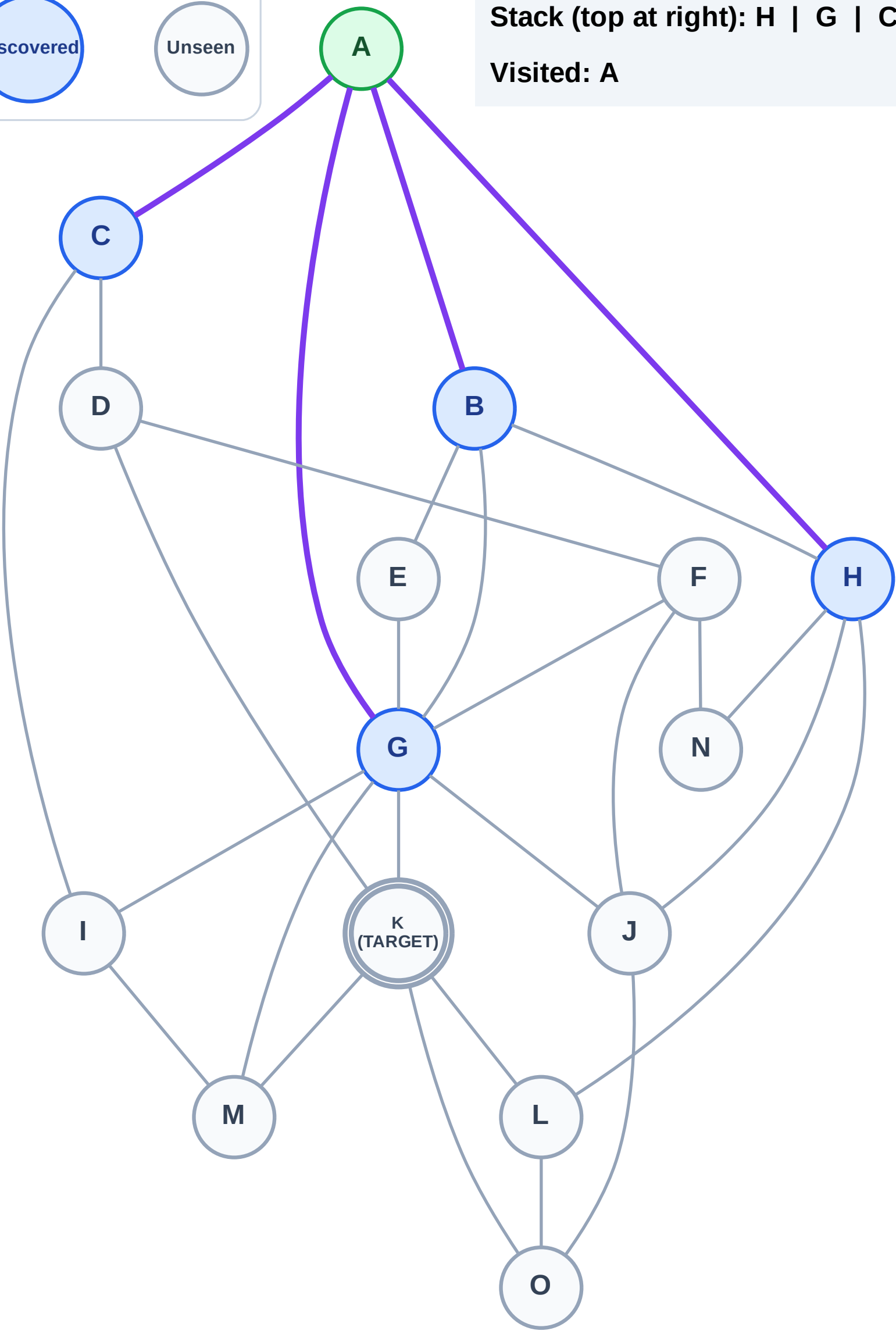
Step 3: Discover H, G, C, B from A

Legend



Stack (top at right): H | G | C | B

Visited: A

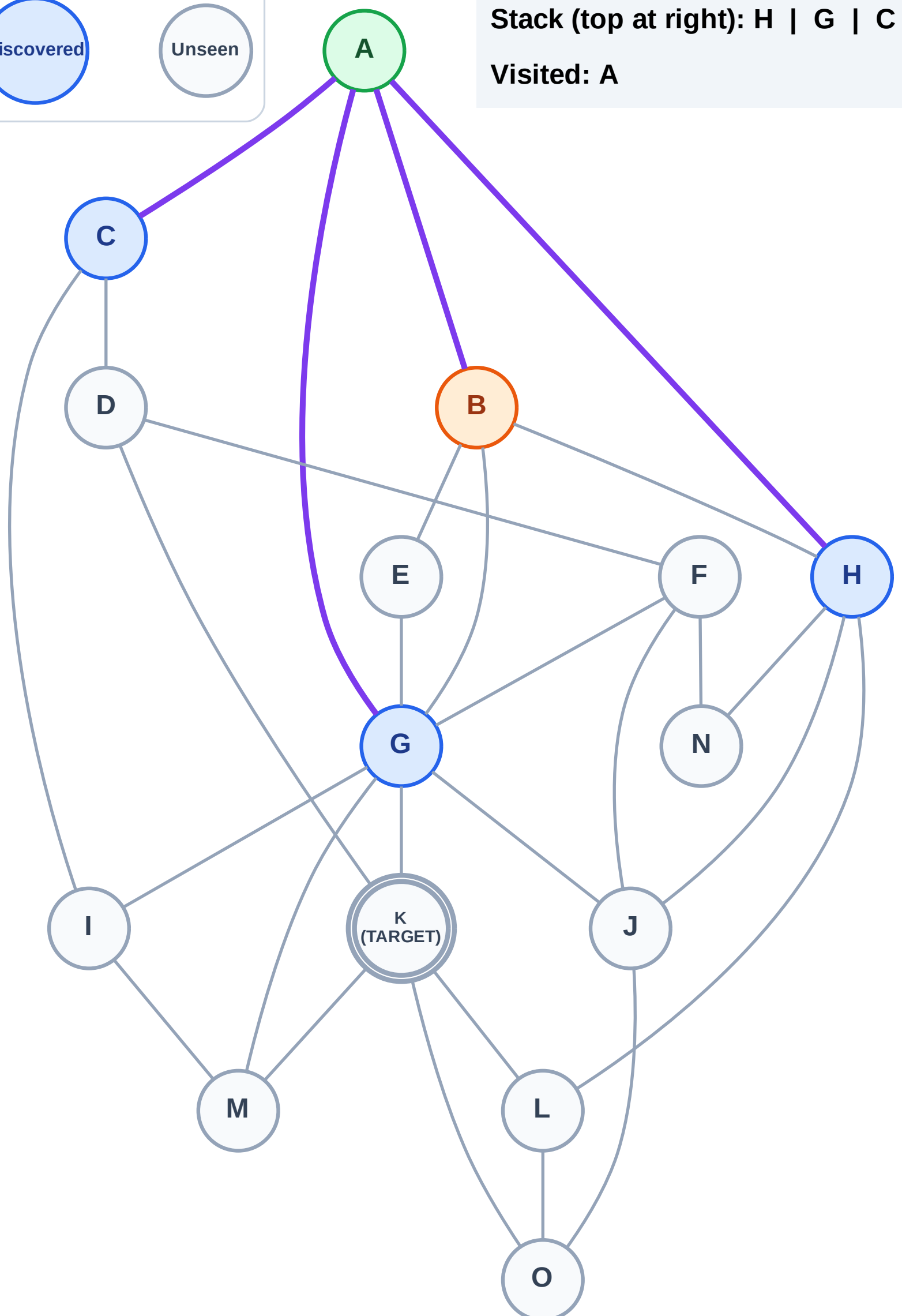


Step 4: Process node B

Legend

Unseen

Visited: A



DFS Traversal

Step 5: Discover E from B

Legend

Complete

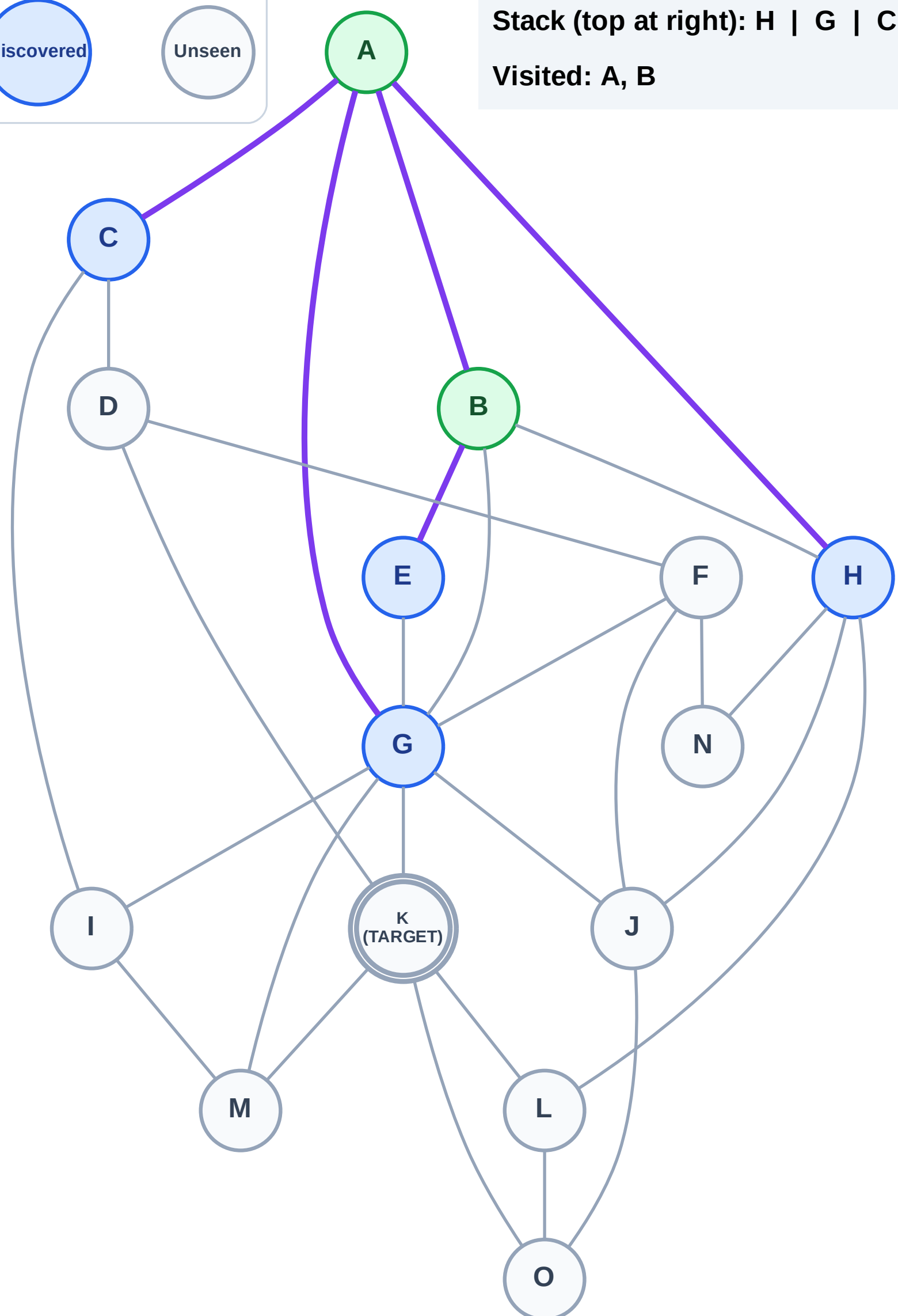
Processing

Discovered

Unseen

Stack (top at right): H | G | C | E

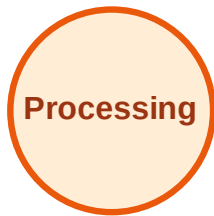
Visited: A, B



DFS Traversal

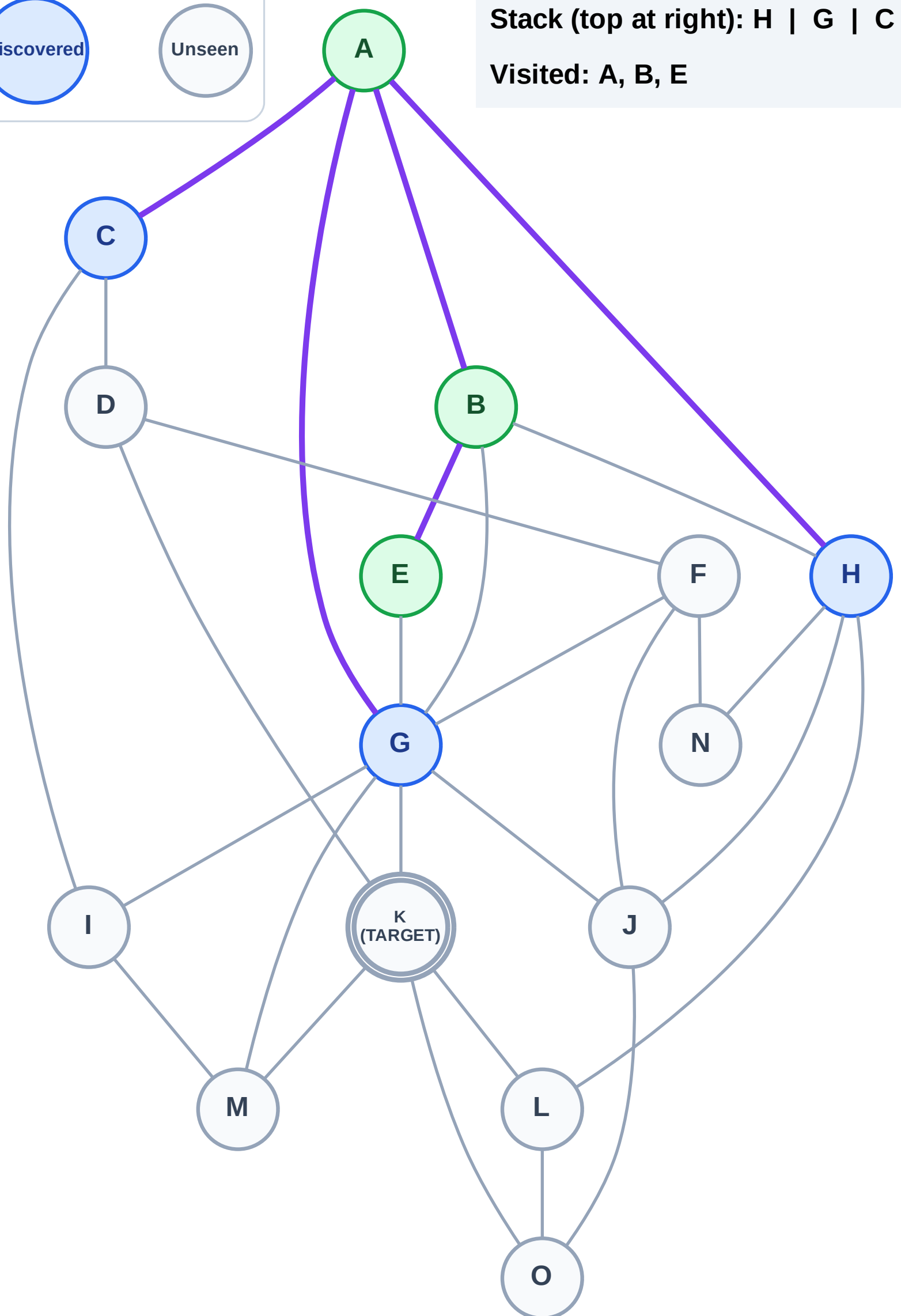
Step 7: No new neighbors from E

Legend



Stack (top at right): H | G | C

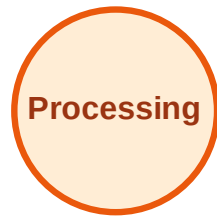
Visited: A, B, E



DFS Traversal

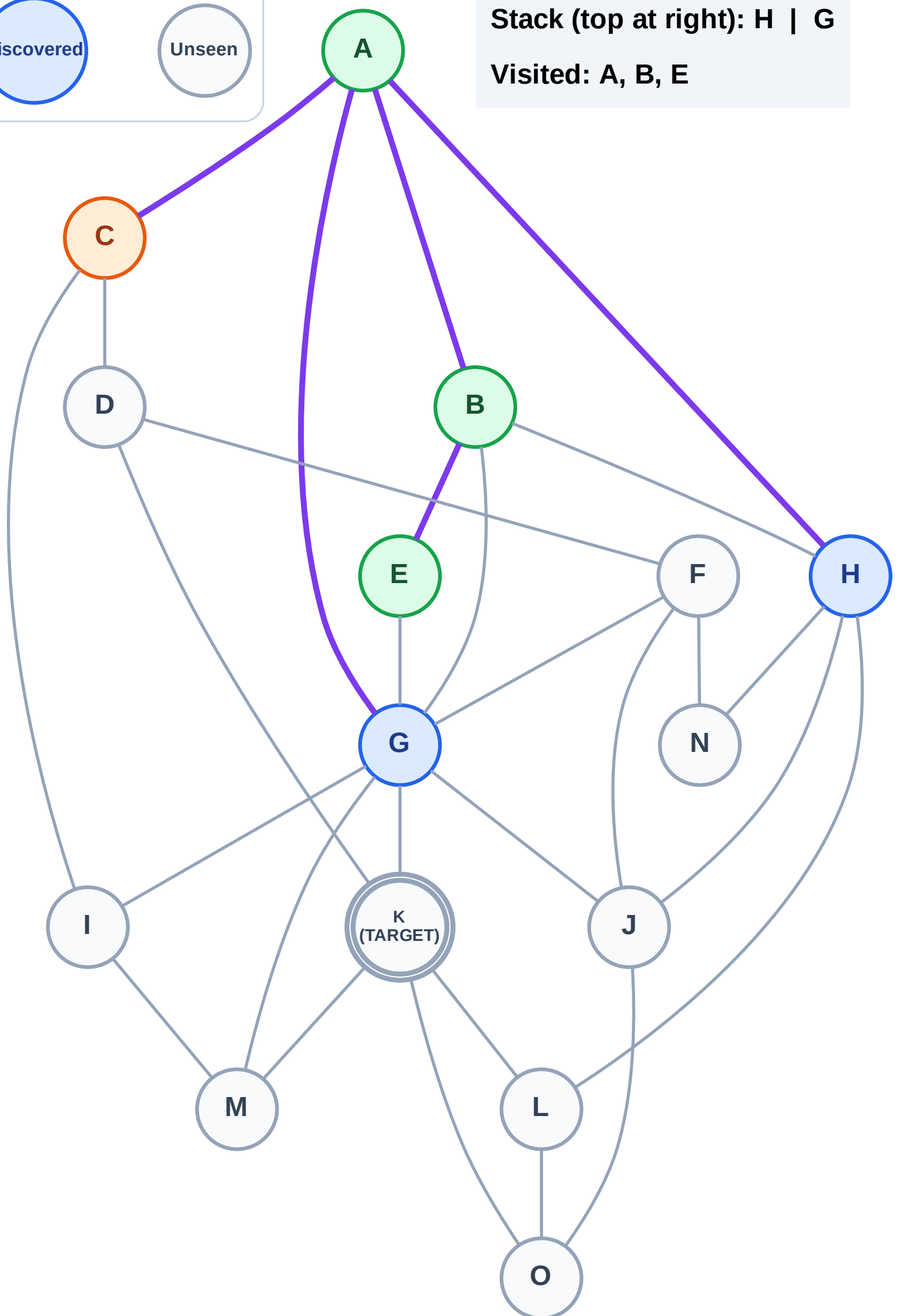
Step 8: Process node C

Legend



Stack (top at right): H | G

Visited: A, B, E

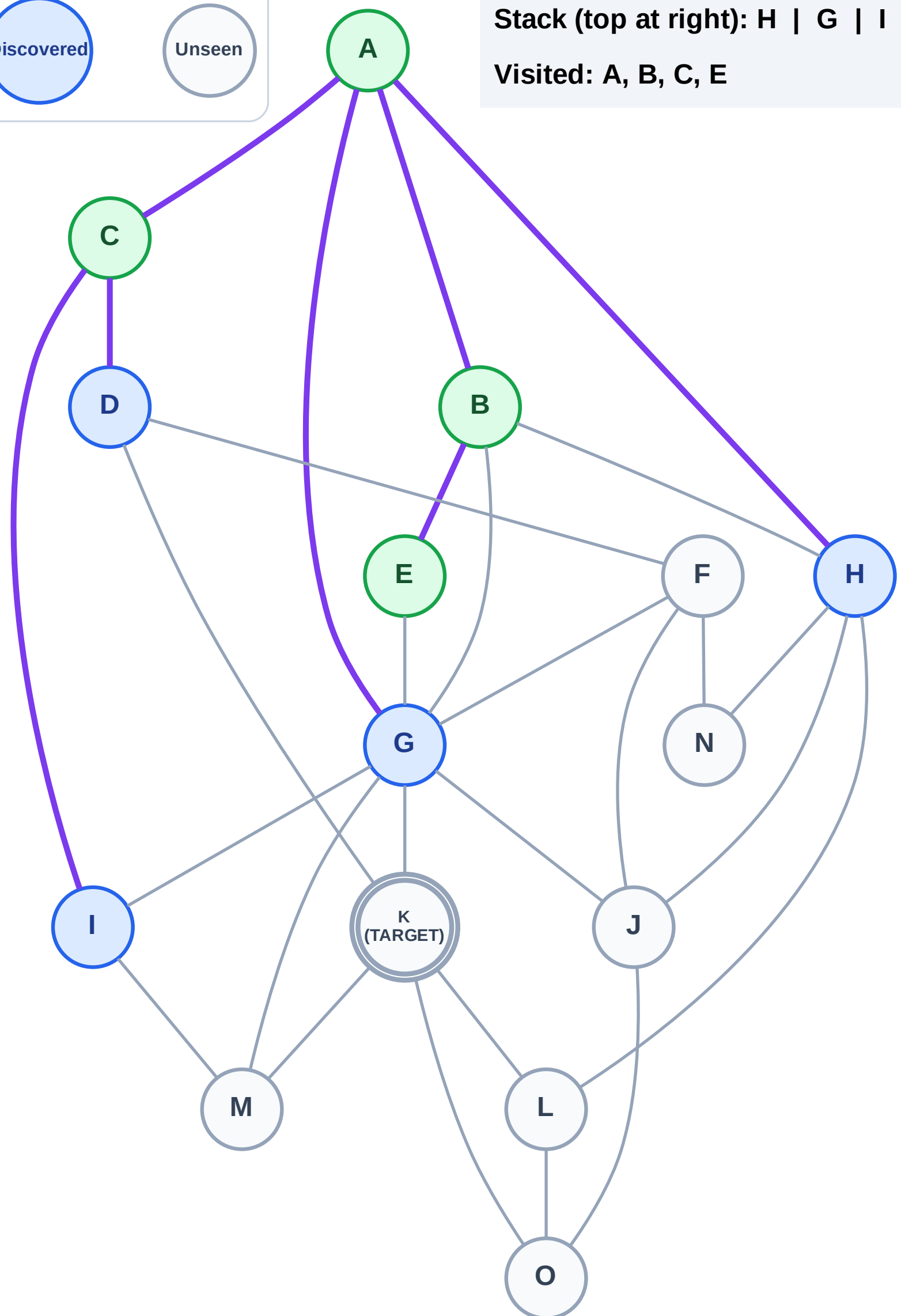


Step 9: Discover I, D from C

Legend

Unseen

Visited: A, B, C, E



DFS Traversal

Step 10: Process node D

Legend

Complete

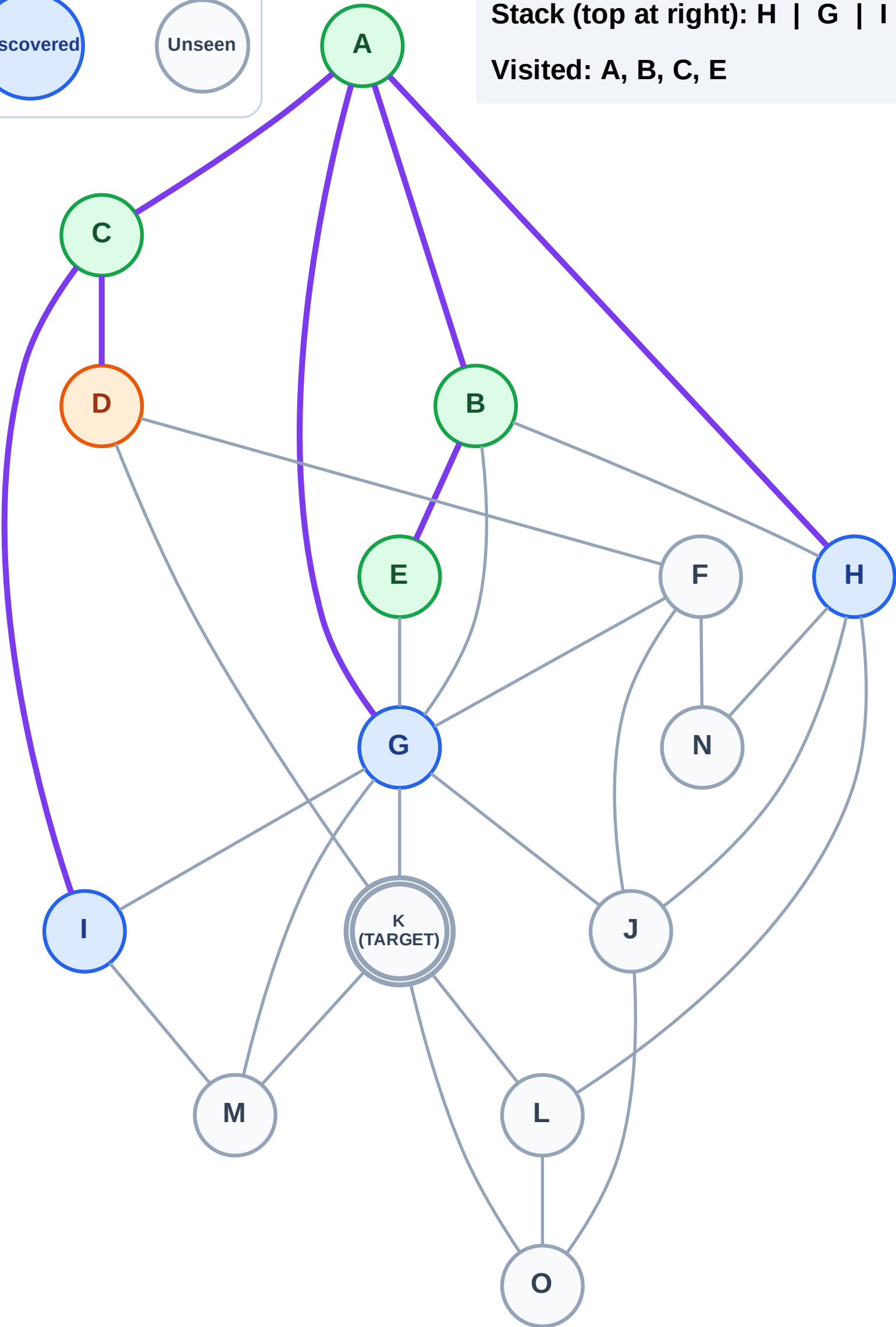
Processing

Discovered

Unseen

Stack (top at right): H | G | I

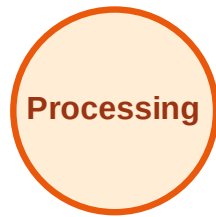
Visited: A, B, C, E



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → C → D → K

Legend



Stack (top at right): H | G | I

Visited: A, B, C, D, E, K

